

ASTR 400B Homework 6 Answers

1. How many close encounters will the MW and M31 experience in the future?

How I'm defining close encounters(which may be wrong) is an encounter where they did not stay together afterwards/passed each other. From the plots I made, there seemed to be two main close encounters where afterward they shifted away, after those two the COMs of both galaxies stayed fairly close together.

2. How is the time evolution of the separation and relative velocity related?

They have almost an inverse relationship. When the two galaxies are close to one another the velocities increase, when they galaxies then get further away, the velocities decrease. Until at the very end when both the position and velocities flatten out.

3. When do M31 and the MW merge? (you might need to zoom in on the plot - try a log y axis). What happens to M33's orbit when they merge?

From my position plot with a log y axis it looks like they begin to merge a little after 6 Gyr but really are merged after 8 Gyr.